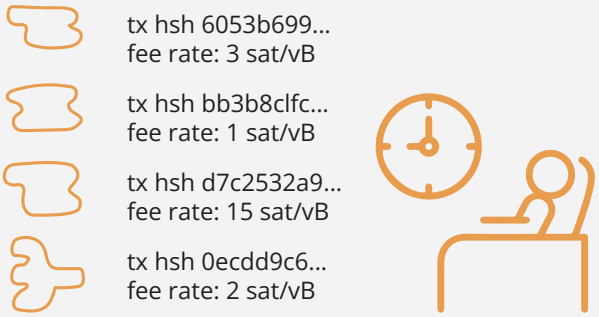
7.3 The "Mempool" or Memory Pool: Understanding the Holding Tank of Bitcoin Transactions

What is the Mempool?

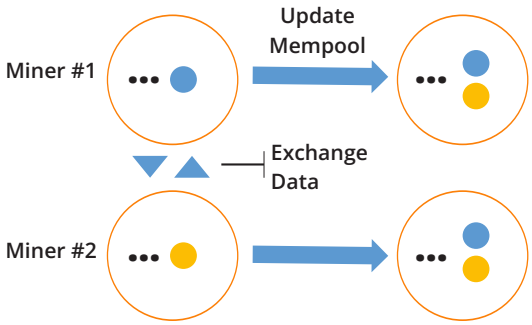
The mempool is like a waiting room for transactions in the ***Bitcoin Network***. When a transaction is made, it is first added to a node's mempool before it is verified and added to the *blockchain*.

A **mempool** is where transactions wait to be confirmed into a block.



When a node fist receives a transaction from a peer, it has to verify the transaction is legit. Nobody wants faulty or deceptive transactions.

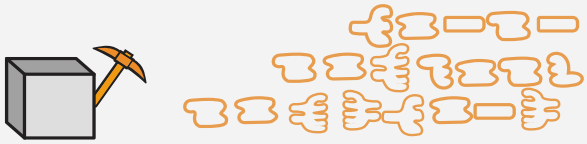
Mempool synchronization allows nodes to share their transactions with other nodes by sending a message containing a list of **verified** transactions in the mempool.



The main purpose of a **mempool** is to:

1 Relay unconfirmed transactions.

2 Provide miners transactions to mine.



Accept To Memory Pool (ATMP) involves checking things like:

- Do I already have this transaction?
- Is there a conflict with a different transaction in the mempool?
- Does the ***bitcoin*** in cover the ***bitcoin*** out?
- Do the signatures prove the previous outputs can be spent?
- Are there enough fees?